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Street Brawl

Ernst Solvay wanted to leave his mark on the world with his money. Like Alfred Nobel before him, he had profited through industrial applications of chemistry—though less explosively than Nobel, the father of dynamite—and, also like Nobel, he wanted to better the world by promoting scientific research. So, in 1911, Solvay used his money to organize a conference on the nascent quantum theory in his native Belgium. The conference was an enormous success, and Solvay decided to pour more money into organizing invitation-only conferences on subjects at the cutting edge of physics and chemistry. Solvay himself died in 1922, but his conferences continue to this day and are among the most rarefied of all scientific meetings. Yet the Fifth Solvay Conference, held in Brussels in October 1927, stands out from the rest. Seventeen of the twenty-nine attendees had won or would go on to win Nobel Prizes; one person there, Marie Curie, already had two Nobels to her name. In addition to Curie, Einstein, Planck, Schrödinger, Bohr, Heisenberg, Born, Dirac, and Pauli were there, and the conference photograph appears in many quantum physics textbooks. And, along with this picture, a historical fable has been handed down from generation to generation of physicists through an informal oral tradition, a sort of origin myth for quantum physics itself, that goes like this:

Once upon a time, a group of brilliant physicists discovered quantum physics. The new theory was wildly successful. Yet Einstein couldn’t accept the radical new picture of nature revealed by quantum physics, despite the pivotal role he had played in its early development (and despite the fact that